

Practice Set 6

MA628: Financial Derivatives Pricing

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1. Let $\Omega = \{1, 2, 3\}$. Write all the possible σ -algebras of Ω .
2. Let $(\Omega, \mathcal{F})$ be a sample space with σ -algebra $\mathcal{F}$. Show that if $A_1, A_2, \dots, A_n \in \mathcal{F}$, then $A_1 \cap A_2 \cap \dots \cap A_n$ also belongs to $\mathcal{F}$.
3. Let $\Omega = [0, 1]$. Find the σ -algebra generated by the events $[0, \frac{1}{3})$ and $[\frac{1}{3}, 1]$.
4. If $\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_n$ are σ -algebras. Show that their intersection is also a σ -algebra.
5. Give examples of two σ -algebras $\mathcal{F}_1$ and $\mathcal{F}_2$ such that $\mathcal{F}_1 \cup \mathcal{F}_2$ is not a σ -algebra.
6. Let $S_1, S_2, \dots, S_n$ be a partition of the sample space Ω (i.e. $\cup_{i=1}^n S_i = \Omega, S_i \cap S_j = \emptyset, i \neq j$). Show that for any event $A, \mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(S_i \cap A)$.
7. Let $\Omega = \{1, 2, 3, 4\}$ and $\mathcal{F} = \{\emptyset, \Omega, \{1, 2\}, \{3, 4\}\}$. Assume that $\mathbb{P}(\{1, 2\}) = \frac{5}{8}$ and $\mathbb{P}(\{3, 4\}) = \frac{5}{8}$. Is $\mathbb{P}$ a probability measure on $(\Omega, \mathcal{F})$?
8. Let $\Omega = \{1, 2, 3, 4\}$. Is any of following collection of sets is a σ -field ? Justify your answer
 - (a) $\mathcal{F}_1 = \{\emptyset, \{1, 2\}, \{3, 4\}\}$
 - (b) $\mathcal{F}_2 = \{\emptyset, \Omega, \{1\}, \{2, 3, 4\}, \{1, 2\}, \{3, 4\}\}$
 - (c) $\mathcal{F}_3 = \{\emptyset, \Omega, \{1\}, \{2\}, \{1, 2\}, \{3, 4\}, \{2, 3, 4\}, \{1, 3, 4\}\}$.
9. Find α such that $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{F})$, where

$$\Omega = \{1, 2, 3, 4\} \text{ and } \mathcal{F} = \{\emptyset, \Omega, \{1\}, \{2, 3\}, \{4\}, \{1, 2, 3\}, \{2, 3, 4\}, \{1, 4\}\},$$

and $\mathbb{P}(\{1\}) = \frac{1}{4}, \mathbb{P}(\{2, 3\}) = \frac{1}{2}, \mathbb{P}(\{4\}) = \alpha$. Further compute $\mathbb{P}(\{1, 2, 3\}), \mathbb{P}(\{2, 3, 4\})$ and $\mathbb{P}(\{1, 4\})$.